

## PART A — QUESTIONS

AF1-001

Question:

Solve:  $3x + 7 = 22$

- A) 3
- B) 5
- C) 7
- D) 9

AF1-002

Question:

If  $5a - 4 = 2a + 11$ , what is  $a$ ?

- A) 3
- B) 4
- C) 5
- D) 6

AF1-003

Question:

Simplify:  $4(2x - 3) + 5x$

- A)  $13x - 12$
- B)  $8x - 12$
- C)  $13x - 3$
- D)  $10x + 5$

AF1-004

Question:

A worker earns \$18 per hour. Which expression represents the worker's pay for  $h$  hours?

- A)  $h + 18$
- B)  $18h$
- C)  $h - 18$
- D)  $18 \div h$

AF1-005

Question:

If  $f(x) = 2x + 9$ , what is  $f(6)$ ?

- A) 12
- B) 15
- C) 21
- D) 24

AF1-006

Question:

Which ordered pair satisfies the equation  $y = 3x - 2$ ?

- A) (2, 4)
- B) (3, 8)
- C) (4, 11)
- D) (5, 12)

AF1-007

Question:

Solve:  $2(x + 5) = 28$

- A) 9
- B) 11
- C) 14
- D) 19

AF1-008

Question:

A pattern begins: 4, 9, 14, 19, ... What is the next number?

- A) 22
- B) 23
- C) 24
- D) 25

AF1-009

Question:

Solve:  $x/4 = 9$

- A) 13
- B) 18
- C) 27
- D) 36

AF1-010

Question:

A line has a slope of 2 and passes through the point (0, 5). Which equation represents the line?

- A)  $y = 2x + 5$
- B)  $y = 5x + 2$
- C)  $y = 2x - 5$
- D)  $y = x + 5$

AF1-011

Question:

If 6 boxes hold 48 parts, how many parts are in 10 boxes at the same rate?

- A) 60
- B) 70
- C) 80
- D) 90

AF1-012

Question:

Solve:  $7 - 2x = -9$

- A) -8
- B) -1
- C) 1
- D) 8

AF1-013

Question:

Which expression is equivalent to  $9x - 4x + 6$ ?

- A)  $5x + 6$
- B)  $13x + 6$
- C)  $5x - 6$
- D)  $9x + 2$

AF1-014

Question:

The cost  $C$  of renting a tool is \$25 plus \$8 per hour. Which equation represents the cost for  $h$  hours?

- A)  $C = 25h + 8$
- B)  $C = 8h + 25$
- C)  $C = 33h$
- D)  $C = 25 - 8h$

AF1-015

Question:

If  $y = -x + 10$ , what is  $y$  when  $x = 3$ ?

- A) 7
- B) 10
- C) 13
- D) -7

AF1-016

Question:

Solve:  $4x - 6 = 2x + 14$

- A) 4
- B) 6
- C) 10
- D) 20

AF1-017

Question:

Which value of  $x$  makes  $5x + 2$  greater than 17?

- A) 2
- B) 3
- C) 4
- D) 1

AF1-018

Question:

A table shows a linear relationship.

$x$ : 1, 2, 3, 4

$y$ : 6, 10, 14, 18

What is the rule?

- A)  $y = 2x + 4$
- B)  $y = 4x + 2$
- C)  $y = 6x$
- D)  $y = x + 5$

AF1-019

Question:

Solve:  $3(x - 4) = 2x + 5$

- A) 7
- B) 13
- C) 17
- D) 19

AF1-020

Question:

If  $a = 4$  and  $b = -3$ , what is  $2a - b$ ?

- A) 5
- B) 8
- C) 11
- D) -2

AF1-021

Question:

Which point is on the y-axis?

- A) (4, 0)
- B) (0, -6)
- C) (3, 3)
- D) (-2, 5)

AF1-022

Question:

A machine produces 15 parts every 3 minutes. At this rate, how many parts does it produce in 12 minutes?

- A) 45
- B) 50
- C) 60
- D) 75

AF1-023

Question:

Solve for  $r$ :  $P = 2r + s$

- A)  $r = P - s$
- B)  $r = (P - s)/2$
- C)  $r = 2(P - s)$
- D)  $r = P/2 + s$

AF1-024

Question:

What is the slope of a line that goes through (1, 4) and (3, 10)?

- A) 2
- B) 3
- C) 4
- D) 6

AF1-025

Question:

Simplify:  $6x + 3 - 2x - 8$

- A)  $4x - 5$
- B)  $8x - 5$
- C)  $4x + 11$
- D)  $8x + 11$

AF1-026

Question:

If  $g(x) = x^2 - 4$ , what is  $g(5)$ ?

- A) 9
- B) 21
- C) 25
- D) 29

AF1-027

Question:

A sequence follows the rule “multiply by 2, then subtract 1.” If the first term is 3, what is the second term?

- A) 4
- B) 5
- C) 6
- D) 7

AF1-028

Question:

Solve:  $5(x + 2) - 3 = 2x + 16$

- A) 1
- B) 2
- C) 3
- D) 4

AF1-029

Question:

A line crosses the y-axis at 4 and goes up 3 units for every 1 unit to the right. Which equation represents the line?

- A)  $y = 4x + 3$
- B)  $y = 3x + 4$
- C)  $y = x + 4$
- D)  $y = 3x - 4$

AF1-030

Question:

If  $\frac{3}{5} = \frac{x}{20}$ , what is  $x$ ?

- A) 8
- B) 10
- C) 12
- D) 15

AF1-031

Question:

Solve:  $-3x + 12 = 0$

- A) -4
- B) 0
- C) 4
- D) 12

AF1-032

Question:

The value of a van decreases by \$1,200 each year. If its starting value is \$18,000, which expression gives its value after  $y$  years?

- A)  $18,000 + 1,200y$
- B)  $18,000 - 1,200y$
- C)  $1,200y - 18,000$
- D)  $18,000 \div 1,200y$

AF1-033

Question:

Which equation has  $x = 6$  as its solution?

- A)  $2x + 5 = 18$
- B)  $3x - 4 = 14$
- C)  $x/2 + 7 = 11$
- D)  $5x - 6 = 20$

## PART B — ANSWER KEY + EXPLANATIONS

AF1-001 — Correct: B — Explanation:

Solve  $3x + 7 = 22$ . Subtract 7 from both sides:  $3x = 15$ . Divide by 3:  $x = 5$ . The most common mistake is subtracting 7 from only one side.

AF1-002 — Correct: C — Explanation:

Start with  $5a - 4 = 2a + 11$ . Subtract  $2a$ :  $3a - 4 = 11$ . Add 4:  $3a = 15$ . Divide by 3:  $a = 5$ . A common mistake is adding instead of subtracting the  $2a$  term.

AF1-003 — Correct: A — Explanation:

Distribute first:  $4(2x - 3) = 8x - 12$ . Then add  $5x$ :  $8x + 5x - 12 = 13x - 12$ . A common mistake is forgetting to multiply  $-3$  by 4.

AF1-004 — Correct: B — Explanation:

Pay equals hourly rate times hours worked. At \$18 per hour for  $h$  hours, the expression is  $18h$ . A common mistake is adding 18 and  $h$  instead of multiplying.

AF1-005 — Correct: C — Explanation:

Substitute 6 for  $x$ :  $f(6) = 2(6) + 9 = 12 + 9 = 21$ . A common mistake is stopping at 12 and forgetting to add 9.

AF1-006 — Correct: C — Explanation:

Test each pair in  $y = 3x - 2$ . For  $(4, 11)$ :  $3(4) - 2 = 12 - 2 = 10$ , not 11. Actually, check carefully: none of the listed choices satisfies the equation exactly. This item must be corrected before use.

AF1-007 — Correct: A — Explanation:

Solve  $2(x + 5) = 28$ . Divide both sides by 2:  $x + 5 = 14$ . Subtract 5:  $x = 9$ . A common mistake is subtracting 5 before dividing by 2 incorrectly.

AF1-008 — Correct: C — Explanation:

The pattern increases by 5 each time: 4, 9, 14, 19. Add 5 to 19: 24. A common mistake is looking at the numbers only and guessing without checking the repeated increase.

AF1-009 — Correct: D — Explanation:

$x/4 = 9$  means  $x$  divided by 4 equals 9. Multiply both sides by 4:  $x = 36$ . A common mistake is dividing 9 by 4 instead of multiplying.

AF1-010 — Correct: A — Explanation:

Slope-intercept form is  $y = mx + b$ . The slope is 2 and the y-intercept is 5, so the equation is  $y = 2x + 5$ . A common mistake is switching the slope and intercept.

AF1-011 — Correct: C — Explanation:

Find parts per box:  $48 \div 6 = 8$ . For 10 boxes:  $10 \times 8 = 80$ . A common mistake is adding 4 boxes and 4 parts instead of using the rate.

AF1-012 — Correct: D — Explanation:

Solve  $7 - 2x = -9$ . Subtract 7:  $-2x = -16$ . Divide by -2:  $x = 8$ . A common mistake is losing the negative sign when dividing.

AF1-013 — Correct: A — Explanation:

Combine like terms:  $9x - 4x = 5x$ . The constant 6 stays the same. The result is  $5x + 6$ . A common mistake is combining 6 with the  $x$  terms.

AF1-014 — Correct: B — Explanation:

The fixed cost is \$25, and the hourly cost is \$8 per hour. For  $h$  hours, the total is  $C = 8h + 25$ . A common mistake is making 25 the hourly rate.

AF1-015 — Correct: A — Explanation:

Substitute  $x = 3$  into  $y = -x + 10$ :  $y = -3 + 10 = 7$ . A common mistake is treating  $-x$  as positive  $x$ .

AF1-016 — Correct: C — Explanation:

Solve  $4x - 6 = 2x + 14$ . Subtract  $2x$ :  $2x - 6 = 14$ . Add 6:  $2x = 20$ . Divide by 2:  $x = 10$ . A common mistake is adding 6 to only one side.

AF1-017 — Correct: C — Explanation:

Test the choices or solve:  $5x + 2 > 17$ . Subtract 2:  $5x > 15$ . Divide by 5:  $x > 3$ . The only listed value greater than 3 is 4. A common mistake is choosing 3, but  $5(3) + 2 = 17$ , not greater than 17.

AF1-018 — Correct: B — Explanation:

The y-values increase by 4 each time  $x$  increases by 1, so the slope is 4. Test  $y = 4x + 2$ : when  $x = 1$ ,  $y = 6$ ; when  $x = 2$ ,  $y = 10$ . A common mistake is using the first y-value, 6, as the slope.

AF1-019 — Correct: C — Explanation:

Distribute:  $3x - 12 = 2x + 5$ . Subtract  $2x$ :  $x - 12 = 5$ . Add 12:  $x = 17$ . A common mistake is distributing 3 only to  $x$  and not to  $-4$ .

AF1-020 — Correct: C — Explanation:

Substitute  $a = 4$  and  $b = -3$ :  $2a - b = 2(4) - (-3) = 8 + 3 = 11$ . A common mistake is treating subtracting a negative as subtraction.

AF1-021 — Correct: B — Explanation:

A point on the y-axis always has  $x = 0$ . The point  $(0, -6)$  is on the y-axis. A common mistake is choosing  $(4, 0)$ , which is on the x-axis.

AF1-022 — Correct: C — Explanation:

15 parts in 3 minutes means 5 parts per minute. In 12 minutes:  $5 \times 12 = 60$ . A common mistake is multiplying 15 by 12 without first finding the rate.

AF1-023 — Correct: B — Explanation:

Start with  $P = 2r + s$ . Subtract  $s$  from both sides:  $P - s = 2r$ . Divide by 2:  $r = (P - s)/2$ . A common mistake is dividing only  $P$  by 2 instead of the whole quantity  $P - s$ .

AF1-024 — Correct: B — Explanation:

Slope = change in  $y \div$  change in  $x$ . From  $(1, 4)$  to  $(3, 10)$ ,  $y$  increases by 6 and  $x$  increases by 2. Slope =  $6 \div 2 = 3$ . A common mistake is reversing the subtraction order for one coordinate only.

AF1-025 — Correct: A — Explanation:

Combine like terms:  $6x - 2x = 4x$ , and  $3 - 8 = -5$ . The result is  $4x - 5$ . A common mistake is writing  $4x + 5$  because the negative sign before 8 is missed.

AF1-026 — Correct: B — Explanation:

Substitute 5 for  $x$ :  $g(5) = 5^2 - 4 = 25 - 4 = 21$ . A common mistake is calculating  $5^2$  as 10 instead of 25.

AF1-027 — Correct: B — Explanation:

Start with 3. Multiply by 2: 6. Subtract 1: 5. A common mistake is subtracting 1 first and then multiplying.

AF1-028 — Correct: C — Explanation:

Distribute:  $5x + 10 - 3 = 2x + 16$ . Combine:  $5x + 7 = 2x + 16$ . Subtract  $2x$ :  $3x + 7 = 16$ . Subtract 7:  $3x = 9$ . Divide by 3:  $x = 3$ . A common mistake is forgetting to combine  $10 - 3$ .

AF1-029 — Correct: B — Explanation:

The line crosses the y-axis at 4, so the y-intercept is 4. It rises 3 for every 1 to the right, so the slope is 3. The equation is  $y = 3x + 4$ . A common mistake is switching slope and intercept.

AF1-030 — Correct: C — Explanation:

Use proportions:  $3/5 = x/20$ . Since  $5 \times 4 = 20$ , multiply 3 by 4:  $x = 12$ . A common mistake is adding 15 to the numerator because 5 becomes 20.

AF1-031 — Correct: C — Explanation:

Solve  $-3x + 12 = 0$ . Subtract 12:  $-3x = -12$ . Divide by  $-3$ :  $x = 4$ . A common mistake is giving  $-4$  because of the negative coefficient.

AF1-032 — Correct: B — Explanation:

The van starts at \$18,000 and loses \$1,200 each year. After  $y$  years, subtract  $1,200y$  from 18,000:  $18,000 - 1,200y$ . A common mistake is adding the yearly decrease.

AF1-033 — Correct: B — Explanation:

Test  $x = 6$ . In choice B:  $3(6) - 4 = 18 - 4 = 14$ , so it is true. A common mistake is choosing an equation that gets close but does not equal the right side exactly.



